E_1 PRIMACY IN MODULAR KOSZUL DUALITY: THE ORDERED BAR COMPLEX, THE AVERAGING MAP, AND FIVE INVARIANTS SURVIVING A Σ_n-QUOTIENT

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Abstract. For a cyclic chiral algebra $\mathcal{A}$, the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct is the level-2 chain object before averaging. The symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$, which underlies the classical Francis–Gaitsgory factorization picture, is its R-twisted coinvariant image when descent exists; in the pole-free locus this is the ordinary Σ_n-coinvariant image under an admissible averaging comparison $\mathrm{av}\colon \mathfrak{g}^{E_1}_{\mathcal{A}} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$. The scalar modular characteristic $\kappa(\mathcal{A})$ is recovered at degree 2 from a meromorphic r-matrix $r(z)$ that lives natively on the ordered side: for abelian families the average is κ, while for non-abelian affine Kac–Moody it is the double-pole term κ_{dp}, and the full κ adds the Sugawara shift $\dim(\mathfrak{g})/2$. The Drinfeld associator has a degree-3 coinvariant shadow, while the full associator remains ordered data. The five foundational theorems A–D and H of modular Koszul duality have ordered formulations on the Koszul/descent locus; their Σ_n-shadows recover the symmetric theorems where the descent comparison is available. Information lost under av includes the full braiding, associator-dependent choices of splitting, and cross-degree terms invisible to scalar invariants. Three distinct coalgebra structures feature and must not be conflated: the Harrison coLie structure with $n!/n$ terms at degree n, the coshuffle structure with 2^n terms, and the deconcatenation structure with $n+1$ terms. Only deconcatenation equips the ordered bar. There are three tiers of r-matrix input: the pole-free tier with Koszul-signed $R = \tau$, the vertex-algebra tier where R is read from the local OPE, and the genuinely E_1 tier containing Yangians and Etingof–Kazhdan quantum vertex algebras.

The dimensional uplift from a 2-real-dimensional chiral algebra on a curve to a 3-real-dimensional holomorphic–topological theory is governed by the chiral Deligne–Tamarkin theorem (Theorem 8.1 below): the universal local Swiss-cheese pair $(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ is terminal in the category of two-coloured $\mathsf{SC}^{\mathrm{ch,top}}$-pairs over $\mathcal{A}$, the open colour $\mathcal{A}$ is acted on by the closed colour through the chiral Swiss-cheese mixed operations, and the closed sector is not acted on by the open sector. The bar coalgebra $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ is a chain-level computational model that resolves $\mathcal{A}$ for the purpose of computing $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ via $\mathrm{Hom}_{\mathrm{ch}}(\overline{B}^{\mathrm{ord}}(\mathcal{A}), \mathcal{A})$; it does not by itself supply the closed colour or the dimensional uplift.

1. Introduction

The bar complex attached to an augmented associative algebra admits two very different presentations. In the first, one takes the reduced tensor coalgebra $T^c(s^{-1}\bar{A})$ with its deconcatenation coproduct; the bar differential is built from the associative product in consecutive slots. In the second, one passes to Σ_n-coinvariants and works with the symmetric bar complex, which is the natural home for commutative or E_∞-algebras and for the Francis–Gaitsgory chiral bar. The ordered bar complex is the level-2 chain object before averaging; the symmetric bar is obtained from it by descent and coinvariants. The averaging map is R-twisted when the spectral kernel is nontrivial and ordinary only in the pole-free locus. Under an admissible descent datum it gives a dg Lie comparison from the ordered convolution algebra to the symmetric one. The five foundational theorems of modular Koszul duality (Theorems A through D,

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together with the chiral Hochschild theorem H) have ordered versions on the Koszul/descent locus, and each symmetric theorem is the Σ_n-coinvariant shadow of its ordered counterpart. The information lost under av is geometric, arithmetic, and physical: the spectral r-matrix, the KZ connection, the Drinfeld associator, and the full braiding of tensor categories of modules all live on the ordered side.

Three principles motivate the inversion. The first is operadic: Stasheff's associahedra K_n (cf. [15]) carry more information than the symmetric groups Σ_n, and the Getzler–Jones [9] theory of E_n-operads makes E_1 the lowest rung of a ladder whose every step is non-commutative. The second is representation-theoretic: the classical r-matrix of an affine Kac–Moody algebra at level k is the content of the degree-2 genus-0 Maurer–Cartan element on the ordered side, and its Σ_2-average recovers the degree-2 scalar shadow: for Heisenberg this is κ, while for non-abelian affine Kac–Moody it is $\kappa_{\mathrm{dp}} = k\dim(\mathfrak{g})/(2h^\vee)$, with the full κ obtained after adding the Sugawara shift $\dim(\mathfrak{g})/2$. The third is operadic–holographic: the chiral Deligne–Tamarkin theorem (Theorem 8.1 below) constructs the universal Swiss-cheese pair $(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ as terminal in the category of two-coloured $\mathsf{SC}^{\mathrm{ch,top}}$-pairs over $\mathcal{A}$. The closed colour $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ acts on the open colour $\mathcal{A}$ through the chiral Swiss-cheese mixed operations; the open colour does not act back on the closed colour. The bar coalgebra $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ is a chain-level computational model used to compute and present this Swiss-cheese pair, not the source of the open–closed transition. The averaging map sends the ordered structure to its Σ_n-coinvariant image *within* the open colour; the closed–open Swiss-cheese relation requires the two-coloured operad and is invisible to averaging on a single colour.

The conventions are those of the monograph [12], referred to below as Volume I. OPE-mode notation is used throughout; conformal weights are denoted h and desuspended degrees $|s^{-1}v| = |v| - 1$ (the bar complex lowers degree). The grading is cohomological.

1.1. **Notation and conventions.** Let k be a field of characteristic zero, X a smooth algebraic curve over k, and $\mathcal{A}$ an augmented cyclic chiral algebra on X in the sense of Beilinson–Drinfeld. Write $\bar{\mathcal{A}}$ for the augmentation ideal $\ker(\varepsilon\colon \mathcal{A} \to k)$. All bar constructions are defined on $\bar{\mathcal{A}}$, not on $\mathcal{A}$ itself. Symmetric groups Σ_n act on tensor powers in the standard way. Over characteristic zero, the finite group algebra $k[\Sigma_n]$ is semisimple, so invariants and coinvariants are exact and identified by the Reynolds idempotent. Coinvariants are denoted with a lower subscript, $(-)_{\Sigma_n}$. The ordered and symmetric Fulton–MacPherson compactifications are written $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$ and $\overline{\mathrm{FM}}_n(\mathbb{C})$ respectively; the former is the Ran-space blow-up along ordered collision diagonals.

2. Three coalgebra structures

A sum of tensor powers carries three distinct cooperadic structures. They differ as natural transformations and by their degree-n coproduct term counts. Only deconcatenation equips the ordered bar complex.

Definition 2.1 (The three coalgebra structures on $T(V)$)**.** Let V be a graded vector space and $T(V) = \bigoplus_{n\geq 0} V^{\otimes n}$ its tensor space. Three cooperad structures on $T(V)$ (or on the ordered/symmetric variants) arise naturally:

(i) The *Harrison coLie* (or co-Lie) cooperad Lie^c, whose degree-n component has dimension $(n-1)!$ and whose coproduct decomposes a Lie word into pairs of sub-Lie-words using the Harrison differential.

(ii) The *coshuffle cooperad* Sym^c, whose degree-n coproduct produces 2^n summands by the shuffle formula
$$\Delta_{\mathrm{sh}}(v_1 \cdots v_n) = \sum_{I \sqcup J = \{1,\dots,n\}} \varepsilon(I,J)\,(v_I) \otimes (v_J),$$

with the sum running over all 2^n bipartitions and $\varepsilon(I, J)$ the Koszul sign.

(iii) The *deconcatenation cooperad* T^c, whose degree-n coproduct produces $n + 1$ summands by splitting the word at a single position:

$$\Delta(v_1 \cdots v_n) = \sum_{i=0}^{n} (v_1 \cdots v_i) \otimes (v_{i+1} \cdots v_n).$$

The three structures are not interconvertible without loss of information. At degree 3 the Harrison differential sees 2 terms, the coshuffle sees 8, and deconcatenation sees 4. Only the last is ordered and strictly associative at the coalgebra level.

Proposition 2.2 (Coshuffle is not deconcatenation)**.** *Let V be nonzero. The coshuffle coproduct on* $\mathrm{Sym}(V)$ *and the deconcatenation coproduct on $T^c(V)$ are distinct cooperadic structures: they have different degree-n term counts (2^n vs. $n + 1$), different symmetry properties (the coshuffle is cocommutative, deconcatenation is not), and different primitive subspaces.*

Proof. Cocommutativity of the coshuffle follows from the involution $I \leftrightarrow J$ on bipartitions. Deconcatenation does not admit such an involution: swapping $(v_1 \cdots v_i)$ with $(v_{i+1} \cdots v_n)$ does not preserve the ordered word. The term counts are elementary. □

The three cooperads sit in a commutative square of natural transformations whose top arrow is the symmetrization and whose bottom arrow is the Harrison projection, but they are not canonically isomorphic. Conflating coshuffle with deconcatenation is a standard error in the chiral-algebra literature and produces spurious identifications between factorization coproducts and bar coproducts.

Remark 2.3 (Where each structure appears). The Harrison coLie cooperad Lie^c underlies the chiral Lie bar complex used by Francis–Gaitsgory, where only the zeroth product of the OPE is visible. The coshuffle cooperad Sym^c underlies the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ on the Ran space, where factorization is implemented by the topological coproduct on unordered configuration space. The deconcatenation cooperad T^c underlies the ordered bar $\overline{B}^{\mathrm{ord}}(\mathcal{A})$, where the coproduct splits a word along a single internal edge and remembers the ordering on the collision points. The averaging map av of Section 4 descends from T^c to Sym^c by external Σ_n-coinvariants; it does not factor through Harrison unless one further restricts to the Eulerian weight-1 summand.

3. The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A})$

For a cyclic chiral algebra, the ordered bar complex is the deconcatenation coalgebra on the ordered configuration space.

Definition 3.1 (The ordered bar complex)**.** Let $\mathcal{A}$ be a strongly admissible augmented E_1-chiral algebra on X, with augmentation ideal $\bar{\mathcal{A}} = \ker \varepsilon$. The *ordered bar complex* is the graded vector space

$$\overline{B}^{\mathrm{ord}}(\mathcal{A}) \;:=\; T^c\big(s^{-1}\bar{\mathcal{A}}\big) \;=\; \bigoplus_{n\geq 0} \big(s^{-1}\bar{\mathcal{A}}\big)^{\otimes n}, \tag{3.1}$$

equipped with the deconcatenation coproduct $\Delta([a_1|\cdots|a_n]) = \sum_{i=0}^{n} [a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_n]$ and the bar differential d_{bar} obtained by extracting OPE collision residues at consecutive pairs of points in the ordered configuration space. The augmentation ideal $\bar{\mathcal{A}}$ is essential here: using $\mathcal{A}$ in place of $\bar{\mathcal{A}}$ in (3.1) would include the unit and break the construction.

The desuspension s^{-1} lowers the degree by one, so $|s^{-1}a| = |a| - 1$. The bar differential lowers tensor degree by one; the coproduct is a chain map with respect to d_{bar} by associativity of the chiral product in consecutive slots. Both the differential and the coproduct are built from the *ordered* collision pattern on $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$; the form $\eta = d\log(z_i - z_j)$ is the universal Arnold propagator (weight one, not weight h).

Theorem 3.2 (Ordered bar differential squared). *The ordered bar differential satisfies* $d_{\mathrm{bar}}^2 = 0$.

Proof. The argument is a codimension-two face pairing on $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$ with the Arnold relation supplying the required cancellation between the two ways of contracting an adjacent pair of collisions. The full face-pairing argument is the ordered Fulton–MacPherson compactification calculation of [12, Chapter 3]. □

Remark 3.3 (Three bar complexes, one map). Let $\overline{B}^{\mathrm{FG}}(\mathcal{A})$ denote the Francis–Gaitsgory chiral Lie bar, which uses only the zeroth OPE product $a_{(0)}b$; let $\overline{B}^{\Sigma}(\mathcal{A}) = \overline{B}^{\mathrm{ch}}(\mathcal{A})$ denote the symmetric bar, which uses all OPE products and takes Σ_n-coinvariants (the bar complex of Volume I, Theorem A); and let $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ denote the ordered bar of (3.1). There is a natural surjection

$$\overline{B}^{\mathrm{ord}}(\mathcal{A}) \twoheadrightarrow \overline{B}^{\Sigma}(\mathcal{A}), \qquad \mathrm{gr}\,\overline{B}^{\Sigma}(\mathcal{A}) \twoheadrightarrow \overline{B}^{\mathrm{FG}}(\mathcal{A}),$$

where the first arrow is Σ_n-coinvariant projection and the second is associated graded along the OPE filtration. These three complexes must not be conflated: the Francis–Gaitsgory complex sees only the Lie part, the symmetric bar sees the full symmetric OPE, and the ordered bar sees the braiding as well.

Example 3.4 (Heisenberg ordered bar). Let $\mathcal{H}_k$ be the Heisenberg chiral algebra at level k, generated by a single field J of conformal weight $h = 1$. The desuspended generator has degree $|s^{-1}J| = 0$. At degree 2 the ordered bar component is

$$(s^{-1}\bar{\mathcal{H}}_k)^{\otimes 2} \cong k \cdot [J|J],$$

a one-dimensional space. The bar differential extracts the collision residue of $J(z_1)J(z_2) \sim k/(z_1 - z_2)^2$ against the Arnold form $\eta = d\log(z_1 - z_2)$, producing a single pole $k/(z_1 - z_2)$.

Remark 3.5 (Heisenberg flat and curved R-data). The Heisenberg algebra $\mathcal{H}_k$ has a flat presentation $r(z) = k/z$ and a curved presentation $R(z) = \exp(k\quad/z)$. They are related by $R(z) = \exp(\quad r(z))$, so $r(z)$ is the classical coefficient of $\log R$. Since $r(z)$ is central in the two tensor factors, the exponentiated form satisfies the quantum Yang–Baxter equation order by order in   .

4. The averaging map av

The averaging comparison realises the symmetric bar as the Σ_n-coinvariant shadow of the ordered bar after a descent datum has been chosen. Its domain is the E_1 convolution dg Lie algebra, its codomain is the modular convolution dg Lie algebra of Volume I, and its kernel records the information lost in the passage from the open to the closed sector.

Definition 4.1 (The E_1 modular convolution dg Lie algebra). Let $\mathcal{A}$ be a cyclic E_1-chiral algebra and let $\mathcal{M}_{\mathrm{Ass}}$ denote the ribbon modular operad with $\mathcal{M}_{\mathrm{Ass}}(g,n) = C_*(\overline{\mathcal{M}}_{g,n}^{\mathrm{rib}})$, the chain complex of the moduli of stable Riemann surfaces with n parametrised boundary circles. Define

$$\mathfrak{g}_{\mathcal{A}}^{E_1} := \prod_{\substack{g,n \\ 2g-2+n>0}} \mathrm{Hom}\big(\mathcal{M}_{\mathrm{Ass}}(g,n),\ \mathrm{End}_{\mathcal{A}}(n)\big).$$

The Hom is taken *without* Σ_n-equivariance: this is the structural feature distinguishing $\mathfrak{g}_{\mathcal{A}}^{E_1}$ from the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ of Volume I, in which the Hom is Σ_n-equivariant and the source cooperad is of Sym^c-type; the ordered source is of T^c-type. The dg Lie structure on $\mathfrak{g}_{\mathcal{A}}^{E_1}$ is inherited from the Feynman transform $F\mathrm{Ass}$: the differential contracts internal edges of ribbon graphs, the bracket composes ribbon graphs.

Definition 4.2 (Admissible averaging datum and averaging map)**.** An *admissible averaging datum* is a strong-unitary R-twisted Σ_n-descent action ρ_R on the ordered bar local systems, satisfying the Yang–Baxter relation, $R_{12}(z)R_{21}(-z) = \mathrm{id}$, and Fulton–MacPherson boundary compatibility. Equivalently, after descent it gives chain maps

$$q_{g,n} : \mathcal{M}_{\mathrm{Ass}}(g,n)_{\Sigma_n} \longrightarrow \mathcal{M}_{\mathrm{Com}}(g,n)$$

compatible with the modular-operadic differentials and edge compositions. With such a datum fixed, the *averaging map*

$$\mathrm{av}_R \;:\; \mathfrak{g}^{E_1}_{\mathcal{A}} \;\longrightarrow\; \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}, \qquad \mathrm{av}_R(\phi)(g,n) \;:=\; \big[\phi(g,n)\big]_{R-\Sigma_n}$$

is the induced R-twisted coinvariant comparison after the descent identification. In the pole-free case $R = \mathrm{id}$, this is the ordinary Reynolds quotient. Without this descent datum no map $\mathfrak{g}^{E_1}_{\mathcal{A}} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is asserted.

Theorem 4.3 (Averaging is a dg Lie comparison on the descent locus)**.** *For an admissible averaging datum, the map* $\mathrm{av}_R : \mathfrak{g}^{E_1}_{\mathcal{A}} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ *is a morphism of dg Lie algebras on the descent locus. It commutes with the differential because ribbon edge contraction descends to stable edge contraction under the strong-unitary R-descent, and it commutes with the Lie bracket because the twisted coinvariant comparison is compatible with modular-operadic composition. Aritywise surjectivity holds after choosing linear splittings of the finite descent quotients; no canonical dg Lie section is part of the datum.*

Proof. The hypotheses say exactly that the R-twisted descent quotient intertwines edge-contraction differentials and modular-operadic compositions. Applying $\mathrm{Hom}(-, \mathrm{End}_{\mathcal{A}})$ and passing to the descent quotient gives a chain map preserving the convolution bracket. Over characteristic zero the finite group algebra $k[\Sigma_n]$ is semisimple, so each arity admits linear splittings of the quotient maps. These splittings are not canonical and need not respect the full cross-arity differential. $\square$

Theorem 4.4 (Degree-2 averaging: Heisenberg exact, affine KM with Sugawara shift)**.** *Let* $r(z) \in \mathrm{End}(V \otimes V) \otimes \mathcal{O}(*\Delta)$ *be the genus-0 degree-2 E_1 shadow of $\mathcal{A}$: the degree-2 component of the E_1 Maurer–Cartan element, in the pre-dualisation convention of Volume I (the shadow-archetype column of* [12, Example]*, equivalently the collision residue computed in $\mathcal{A} \otimes \mathcal{A}$ before applying the Koszul pairing). Then*

$$\mathrm{av}_{R,n=2}\big(r(z)\big) \;=\; \begin{cases} \kappa(\mathcal{A}), & \textit{for abelian Heisenberg families,} \\ \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k) := \dfrac{k\ \dim(\mathfrak{g})}{2h^\vee}, & \textit{for non-abelian affine Kac–Moody in the trace-form convention.} \end{cases}$$

In the affine Kac–Moody case the full modular characteristic is

$$\kappa(\widehat{\mathfrak{g}}_k) \;=\; \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k) + \frac{\dim(\mathfrak{g})}{2} \;=\; \frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee},$$

so the extra $\dim(\mathfrak{g})/2$ *is the Sugawara simple-pole shift. Concretely, for the principal examples:*

- *Heisenberg* $\mathcal{H}_k$*:* $r(z) = k/z$, $\kappa(\mathcal{H}_k) = k$.
- *Affine Kac–Moody* $\widehat{\mathfrak{g}}_k$*:* $r(z) = k\Omega_{\mathfrak{g}}/z$ *(the level k survives; at $k = 0$ the r-matrix vanishes identically),* $\mathrm{av}_R(r(z)) = \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k) = k\dim(\mathfrak{g})/(2h^\vee)$*, and* $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$.
- *Virasoro* Vir_c*:* $r(z) = (c/2)/z^3 + 2T/z$ *(cubic plus simple pole, no quartic), and* $\kappa(\mathrm{Vir}_c) = c/2$.

Proof. The degree-2 Maurer–Cartan component is, by definition, the degree-2 E_1 shadow $r(z) \in \mathrm{End}(V^{\otimes 2}) \otimes \mathcal{O}(*\Delta)$ in the pre-dualisation convention. Averaging over Σ_2 projects onto the trivial isotypic component. For abelian Heisenberg families this scalar coincides with the full modular characteristic. For non-abelian affine Kac–Moody in the trace-form convention $r(z) = k\Omega_{\mathfrak{g}}/z$, the averaging sees only the double-pole channel and yields $\kappa_{\mathrm{dp}} = k\dim(\mathfrak{g})/(2h^\vee)$; the simple-pole self-contraction contributes the additional Sugawara term $\dim(\mathfrak{g})/2$, so $\kappa = \kappa_{\mathrm{dp}} + \dim(\mathfrak{g})/2$. The example computations are recorded in [12, Chapter 5]; each formula is verified by the three independent paths of pole-order analysis, the limiting case at $k = 0$ for the trace-form r-matrix, and cross-family consistency against the landscape census. □

Theorem 4.5 (Degree-3 averaging: associator to cubic shadow). *Let $\Phi_{\mathrm{KZ}}(\mathcal{A}) \in \mathrm{End}(V^{\otimes 3})$ be the genus-0 degree-3 Maurer–Cartan component, which for affine Kac–Moody $\widehat{\mathfrak{g}}_k$ is the Drinfeld–Kontsevich KZ associator. Then*

$$\mathrm{av}_{R,n=3}\big(\Phi_{\mathrm{KZ}}(\mathcal{A})\big) \;=\; \mathfrak{C}(\mathcal{A}),$$

where $\mathfrak{C}(\mathcal{A})$ is the cubic shadow of Volume I, equivalently the Σ_3-coinvariant of the associator. The antisymmetric component $[\Omega_{12}, \Omega_{23}]$ lies entirely in $\ker(\mathrm{av})$*: it is antisymmetric under the transposition* $(1 \leftrightarrow 3)$*, so its Σ_3-average vanishes. Only the symmetric products $\Omega_{12}\Omega_{23} + \Omega_{23}\Omega_{12}$ contribute to $\mathfrak{C}(\mathcal{A})$.*

Proof. The Σ_3-isotypic decomposition of Φ_{KZ} in $\mathrm{End}(V^{\otimes 3})$ splits into the trivial, sign, and standard representations. In the pole-free specialization, the Reynolds operator $\mathrm{av} = (1/6)\sum_{\sigma\in\Sigma_3}\sigma$ projects onto the trivial isotypic component. A direct check on $[\Omega_{12}, \Omega_{23}]$ shows it is antisymmetric under $(1 \leftrightarrow 3)$ and therefore has trivial isotypic component zero. The symmetric products $\Omega_{ij}\Omega_{k\ell} + \Omega_{k\ell}\Omega_{ij}$ are the only surviving contributors. The identification of this coinvariant with the cubic shadow is Volume I Theorem D(iii). □

Proposition 4.6 (Splittings and associator dependence). *For a fixed admissible averaging datum, the sequence of graded linear spaces*

$$0 \longrightarrow \ker(\mathrm{av}) \longrightarrow \mathfrak{g}^{E_1}_{\mathcal{A}} \xrightarrow{\mathrm{av}} \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \longrightarrow 0$$

splits aritywise, but it has no canonical splitting compatible with the full Maurer–Cartan problem. At each fixed degree n, the inclusion $\mathrm{End}^{\Sigma_n}(V^{\otimes n}) \hookrightarrow \mathrm{End}(V^{\otimes n})$ *is a Lie subalgebra embedding. The obstruction to a canonical global section is cross-degree: the bar differential reduces degree by one and can carry a chosen representative in* $\ker(\mathrm{av}_{R,n})$ *to a term with nonzero* $\mathrm{av}_{R,n-1}$*-component. On the associator-controlled subproblem, choices of compatible splitting form the usual Drinfeld associator torsor under* GRT_1.

Proof. Maschke semisimplicity gives aritywise linear splittings over a field of characteristic zero. Compatibility with the bar differential is extra structure, because the differential relates different arities. For KZ/Yangian input the remaining choice is the choice of associator; Drinfeld's theorem identifies the set of associators as a GRT_1-torsor [2], and Etingof–Kazhdan quantisation supplies the corresponding quasi-triangular deformation data [3]. □

5. The five theorems as averaging-invariants

The following are conditional all-genus E_1 comparison surfaces. Their genus-zero finite-degree pieces are proved where cited; the all-genus statements assume the strict ordered Mittag–Leffler tower, PBW-completeness on the Koszul locus, strong-unitary R-descent, and completed finite-window convergence. In the verified families, Theorem D identifies the scalar modular characteristic with the degree-2 coinvariant of $r(z)$.

Theorem 5.1 (Theorem A^{E_1}: ordered bar–cobar). (quadrant *Open,* presentation *bar / twisting coalgebra,* level $1 \leftrightarrow 2$ *of the Beilinson tower,* hypothesis package*: strongly admissible augmented cyclic E_1-chiral algebra $\mathcal{A}$ with an admissible R-twisted descent datum.*) *Let $\mathcal{A}$ be a strongly admissible augmented cyclic*

E_1-chiral algebra with an admissible descent datum. The genus-g ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A})(g,n)$ is an $F\mathrm{Ass}$-coalgebra with differential satisfying $d^2 = 0$, where $F\mathrm{Ass}$ is the Feynman transform of the ribbon modular operad. The bar and cobar functors form an adjunction

$$(\Omega^{\mathrm{ch}})^{(g)} \dashv (\overline{B}^{\mathrm{ord}})^{(g)} : \mathsf{Alg}_{F\mathrm{Ass}} \rightleftarrows \mathsf{Coalg}_{F\mathrm{Ass}},$$

and coinvariant projection recovers Theorem A of Volume I.

Theorem 5.2 (Theorem B^{E_1}: ordered inversion on the Koszul locus)**.** (quadrant *Open,* presentation *bar / cobar inversion,* level $1 \to 3$ *of the Beilinson tower,* hypothesis package*: E_1 Koszul locus* (*PBW-completeness on the ordered bar complex*)*, propagated genus by genus via the MC perturbation lemma.*) *Assume $\mathcal{A}$ lies on the E_1 Koszul locus, meaning the PBW-completeness hypothesis of* [12, Chapter 4] *holds for the ordered bar complex. Then at each genus g, the counit*

$$(\Omega^{\mathrm{ch}})^{(g)}\big(\overline{B}^{\mathrm{ord}}(\mathcal{A})(g,\bullet)\big) \xrightarrow{\ \sim\ } \mathcal{A}$$

is a quasi-isomorphism (the inversion $\Omega^{\mathrm{ch}} \circ \overline{B}^{\mathrm{ord}} \simeq \mathrm{id}$ on the Koszul locus, not a Koszul-duality statement), and the E_1 linear Koszul dual $\mathcal{A}^{!,E_1}$ obtained by linear duality on a finite-dimensional model of $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ is an E_1-chiral algebra at each such genus. The scope of "all genera" here is contingent on the PBW-completeness hypothesis being propagated genus by genus via the MC perturbation lemma of [12, Chapter 9].

Theorem 5.3 (Theorem C^{E_1}: conditional ordered complementarity)**.** (quadrant *Open,* presentation *ordered complementarity,* level 5 *per stratum,* hypothesis package*: ordered Koszul / descent locus, finite perfect model for the ordered bar pairing, ordered Hochschild centre $Z^{\mathrm{der},E_1}_{\mathrm{ch}}(\mathcal{A})$ identified with the centre used by the strict ordered tower.*) *Assume $\mathcal{A}$ lies in the ordered Koszul/descent locus, admits a finite perfect model for the ordered bar pairing, and has an ordered Hochschild centre $Z^{\mathrm{der},E_1}_{\mathrm{ch}}(\mathcal{A})$ identified with the centre used by the strict ordered tower. At each stable (g,n) the ordered complementarity comparison is conditional:*

$$Q^{E_1}_{g,n}(\mathcal{A}) + Q^{E_1}_{g,n}(\mathcal{A}^{!,E_1}) \longrightarrow H^*\big(\overline{\mathcal{M}}^{\mathrm{rib}}_{g,n}, Z^{\mathrm{der},E_1}_{\mathrm{ch}}(\mathcal{A})\big),$$

after applying the ordered centre functor and the perfect pairing. Here the centre is a Hochschild object, not the subspace of R-matrix-fixed vectors, and the comparison is not a consequence of averaging alone. The scalar identity $\kappa + \kappa^! = 0$ for Kac–Moody and free-field families (and $\kappa + \kappa^! = 13$ for Virasoro) is the Σ_n-coinvariant image in the verified scalar examples.

Theorem 5.4 (Theorem D^{E_1}: degree-2 package)**.** (quadrant *Open,* presentation *degree-2 shadow tower,* level 4 *on $\overline{\mathcal{M}}_{g,n}$,* hypothesis package*: genus-refined degree-2 projection of $\Theta^{E_1}_{\mathcal{A}}$ exists genus by genus; cyclic admissibility on each stratum; higher-genus line-side R-matrix interpretation requires additional Yangian input* (*open*)*.*) *The formal ordered degree-2 shadow series*

$$R^{E_1,\mathrm{bin}}(z;\ ) = \sum_{g\geq 0} {}^{2g}\, r_g(z)$$

obtained from the genus-refined degree-2 projection of the E_1 Maurer–Cartan element is the E_1 degree-2 characteristic package of $\mathcal{A}$. It is universal, additive, antisymmetric under opposite-duality, and satisfies $\mathrm{av}_R(r_0(z)) = \kappa(\mathcal{A})$ *on abelian families, while for non-abelian affine Kac–Moody one has* $\mathrm{av}_R(r_0(z)) = \kappa_{\mathrm{dp}}(\mathcal{A})$ *and $\kappa(\mathcal{A}) = \kappa_{\mathrm{dp}}(\mathcal{A}) + \dim(\mathfrak{g})/2$. The coinvariant recovery of the scalar curvature from $r(z)$ is the content of Theorem 4.4. A full higher-genus line-side modular R-matrix interpretation requires additional Yangian input and remains open here.*

Theorem 5.5 (Theorem H^{E_1}: ordered chiral Hochschild)**.** (quadrant *Open,* presentation *ordered chiral Hochschild concentration,* level 3 *of the Beilinson tower,* hypothesis package*: $\mathcal{A}$ complete, ordered Koszul,* *admissible R-twisted descent datum; complete $\mathcal{A}$-bimodule M in the same completion as $\mathcal{A}$; chain-level statement requires weight-completed / pro-object / J-adic ambient for class M.) Let $\mathcal{A}$ be complete, ordered Koszul, and equipped with the admissible descent datum of Definition 4.2. For complete bimodules M in the same completion, the genus-g ordered chiral Hochschild homology is computed by the ordered chiral coHochschild complex of the ordered bar coalgebra $C = \overline{B}^{\mathrm{ord}}(\mathcal{A})$. In the uniform-weight case the twisted differential contains the curvature term $\kappa(\mathcal{A})\lambda_g$. At genus $g \geq 2$ with multi-weight field content the scalar formula receives the cross-channel correction $\delta F_g^{\mathrm{cross}}$ of Volume I. The Σ_n-coinvariant recovers the symmetric chiral Hochschild complex of Theorem H.*

Remark 5.6 (Theorem D as the degree-two model). Theorem D is the degree-two model for the E_1 primacy thesis. The scalar κ is one number per family, while its ordered lift is the meromorphic collision function $r(z)$. For Heisenberg, $\mathrm{av}_R(r(z)) = \kappa$; for non-abelian affine Kac–Moody, $\mathrm{av}_R(r(z)) = \kappa_{\mathrm{dp}}$ and the Sugawara channel adds $\dim(\mathfrak{g})/2$. The projection from $r(z)$ to its scalar shadow loses the braiding, quantum-group, and line-operator data.

Remark 5.7 (What averaging forgets). The kernel of av contains several objects of independent interest. At degree 2, $\ker(\mathrm{av})$ is the antisymmetric part of $r(z)$; for bosonic generators of even desuspended degree this vanishes identically, and κ recovers all of $r(z)$. For generators of odd desuspended degree, the kernel is nontrivial and captures the parity-dependent Eulerian weight structure on the bar complex. At degree 3, $\ker(\mathrm{av})$ contains the linearised Drinfeld commutator $[\Omega_{12}, \Omega_{23}]$ and, by the non-splitting proposition, obstructs a canonical reconstruction of the full associator from $\mathfrak{C}$. The GRT_1-torsor attached to associator-controlled splittings is the operadic shadow of Etingof–Kazhdan non-uniqueness of quantisation.

6. Three tiers of R-matrix

The ordered bar complex distinguishes three tiers of input, corresponding to how the degree-2 Maurer–Cartan component $r(z)$ is constructed. The tiers are a refinement of the E_n-hierarchy restricted to the degree-2 slice; only the third is genuinely E_1 in the sense that the Maurer–Cartan element cannot be recovered from the underlying E_∞-chiral algebra.

Tier (a): pole-free, R is the Koszul-signed flip. If $\mathcal{A}$ is pole-free (that is, the OPE $a(z)b(w)$ has no singularity for any $a, b \in \mathcal{A}$), then the degree-2 collision residue vanishes identically and $r(z) = 0$. In this tier the ordered bar complex admits a canonical Σ_n-equivariant structure: the descent datum R_σ on the local system $\mathcal{F}_n^{\mathrm{ord}}$ is the Koszul-signed permutation

$$R_\sigma \;=\; \varepsilon(\sigma) \cdot \tau_\sigma,$$

where τ_σ permutes tensor factors. The ordered and symmetric bar complexes are quasi-isomorphic in this tier. Every genuinely commutative (in the E_∞ sense) example lives here: the trivial chiral algebra, free-field tensor products with vanishing cross-OPE, and the structure sheaf of a point.

Tier (b): vertex algebras, R from local OPE. If $\mathcal{A}$ is a vertex algebra in the Borcherds–Frenkel–Lepowsky–Meurman sense (more generally, any E_∞-chiral algebra with poles in its OPE), then $r(z)$ is determined by the degree-2 residue of the OPE along the Arnold form $d\log(z_1 - z_2)$. All standard families of conformal field theory live here: Heisenberg $\mathcal{H}_k$ with $r(z) = k/z$, affine Kac–Moody $\widehat{\mathfrak{g}}_k$ with $r(z) = k\Omega_{\mathfrak{g}}/z$, Virasoro Vir_c with $r(z) = (c/2)/z^3 + 2T/z$ (cubic and simple poles, no quartic), $\beta\gamma$, bc, W-algebras, Bershadsky–Polyakov. In each case $r(z)$ is fully determined by the underlying E_∞ vertex algebra structure; the E_1 refinement adds no new data at degree 2 beyond what is already visible in the OPE. The averaging map on tier (b) recovers the scalar degree-2 shadow: for Heisenberg

$\mathrm{av}_R(r(z)) = \kappa(\mathcal{A})$, while for non-abelian affine Kac–Moody $\mathrm{av}_R(r(z)) = \kappa_{\mathrm{dp}}(\mathcal{A})$ and the full $\kappa(\mathcal{A})$ adds the Sugawara shift $\dim(\mathfrak{g})/2$; the kernel is nontrivial but reconstructible from $\mathcal{A}$.

Tier (c): genuinely E_1. The third tier is populated by algebras in which $r(z)$ is independent input: the Maurer–Cartan element cannot be recovered from any underlying E_∞ structure. The paradigmatic examples are the Yangians $Y(\mathfrak{g})$ and the Etingof–Kazhdan quantum vertex algebras [4] obtained from quasi-triangular Lie bialgebras. In this tier the degree-2 Maurer–Cartan element is the rational Yang–Baxter r-matrix $r_{\mathrm{YB}}(z)$, and the associator is a full Drinfeld associator. Existing vertex-algebra theory cannot recognise these algebras, and the ordered bar complex is the minimal framework that makes them visible.

Remark 6.1 (E_1 is the atom of the nonlocal regime). The three tiers form a strict hierarchy. Every standard chiral algebra is E_∞ in the sense that its OPE is symmetric in the operator ordering up to the signs from the Arnold form; it is the interpretation as an E_1 algebra that is the refinement. Tier (c) consists of genuinely nonlocal objects, in the precise sense that the Maurer–Cartan element lies outside the image of any E_∞-to-E_1 promotion functor. The error of calling E_∞-chiral algebras "local" and E_1-chiral algebras "non-abelian" obscures the structure. The operative distinction is symmetric versus ordered, or closed versus open. The averaging map of Section 4 erases the ordering.

7. Four functors, four distinct objects

The structural part closes with a table of functors. Four distinct functors act on a cyclic E_1-chiral algebra $\mathcal{A}$, and each produces a different object. Care is required in keeping them apart.

(i) *Bar.* The ordered bar functor $\mathcal{A} \mapsto \overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$. The output is a coalgebra over the Feynman transform $F\mathrm{Ass}$ of the ribbon modular operad.

(ii) *Cobar.* The cobar functor $C \mapsto \Omega^{\mathrm{ch}}(C)$ on $F\mathrm{Ass}$-coalgebras. On the PBW-complete locus of Theorem 5.2, the composite $\Omega^{\mathrm{ch}} \circ \overline{B}^{\mathrm{ord}}$ is quasi-isomorphic to the identity; this is a statement about the bar–cobar adjunction returning $\mathcal{A}$ to itself, and is structurally distinct from the functor that produces the dual algebra.

(iii) *Verdier.* The Verdier-duality functor $\mathcal{A} \mapsto \mathcal{A}^!_\infty$, where $\mathcal{A}^!_\infty$ is the chain-level dual on the Ran space. Verdier duality intertwines the bar functor in the sense of [12, Theorem A]; the strict linear Koszul dual $\mathcal{A}^!$, obtained by taking a finite-dimensional model and applying linear duality, in general differs from $\mathcal{A}^!_\infty$ and the two must be kept distinct.

(iv) *Derived centre.* The chiral Hochschild cochain functor $\mathcal{A} \mapsto Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = \mathrm{HH}^{*,\mathrm{ch}}(\mathcal{A})$. In the open-closed picture this plays the role of the bulk: the centre of the open (line-operator) algebra $\mathcal{A}$ is the closed (bulk) sector. This functor is *not* the bar–cobar composition; it is a separate construction, the subject of Theorem H.

Theorem A of Volume I is the compatibility statement between (i) and (iii): the bar functor intertwines with Verdier duality on the Ran space. Theorem B (its E_1 version, Theorem 5.2 above) is the internal statement about (i) and (ii): on the PBW-complete locus the bar-cobar composite returns $\mathcal{A}$ up to quasi-isomorphism. Theorem H is the statement about (iv): the derived centre coincides with the ordered chiral Hochschild complex of the bar coalgebra. The four objects are four different outputs; the statement "$\overline{B}^{\mathrm{ord}}(\mathcal{A})$ computes the bulk" is category-theoretically ill-formed, because the closed sector is the derived centre, not the bar.

In the three-pillar language of Volume I, the distinction is: bar-cobar adjointness gives inversion; Verdier duality gives $\mathcal{A}^!$; chiral Hochschild cochains give the closed sector. The averaging map av of

Section 4 intertwines the ordered and symmetric versions of the bar functor on the descent locus. The Verdier, cobar, and Hochschild functors require their own compatibility data; they are not formal consequences of averaging alone.

8. The Swiss-cheese / chiral Deligne–Tamarkin mechanism

The bar functor and the averaging map both live on the open colour of the chiral theory. The dimensional uplift to a 3-real-dimensional holomorphic–topological theory is not accomplished by either of them: it is a property of the universal Swiss-cheese pair built by the chiral Deligne–Tamarkin construction. The bar coalgebra $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ is the chain-level computational model used to present that pair.

Theorem 8.1 (Swiss-cheese / chiral Deligne–Tamarkin)**.** (quadrant *Open,* presentation *two-coloured chiral Swiss-cheese pair,* level $1 \to 3$ *of the Beilinson tower,* hypothesis package*: a local A_∞-chiral algebra $\mathcal{A}$ on a tangential log curve (X, D, τ) with completed conformal-weight topology and finite chiral residue at every double-point collision; a two-coloured chiral Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ with chiral colour and topological collar colour* [17, 18]*; an inner conformal vector controlling $\bar{Q}$-translations on the topological collar.*) *The universal local chiral Swiss-cheese pair*

$$U(\mathcal{A}) \;=\; \big(\mathrm{HH}^{*,\mathrm{ch}}(\mathcal{A},\mathcal{A}),\ \mathcal{A},\ \{\mu^{\mathrm{univ}}_{p;q}\}\big) \;=\; \big(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}),\ \mathcal{A},\ \{\mu^{\mathrm{univ}}_{p;q}\}\big)$$

is terminal in the category of local $\mathsf{SC}^{\mathrm{ch,top}}$-pairs over $\mathcal{A}$: the closed colour $\mathrm{HH}^{*,\mathrm{ch}}(\mathcal{A},\mathcal{A}) = Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ acts on the open colour $\mathcal{A}$ through the chiral Swiss-cheese mixed operations $\mu_{p;q}$ (p closed inputs, q open inputs); the open colour does not act back on the closed colour. For any local $\mathsf{SC}^{\mathrm{ch,top}}$-pair $(B, \mathcal{A}, \{\mu_{p;q}\})$ over $\mathcal{A}$ with continuous mixed operations there is a unique morphism $\Phi\colon (B, \mathcal{A}, \mu) \to U(\mathcal{A})$ over the identity on $\mathcal{A}$.

Citation. This is Volume I's chiral Deligne–Tamarkin theorem, the chiral analogue of the Voronov Swiss-cheese formality theorem [17] and the Kontsevich–Soibelman Deligne conjecture [18]. Concretely, $\Phi(b)_n$ is the chiral Hochschild cochain $\Phi(b)_n(a_1, \dots, a_n; \lambda_1, \dots, \lambda_{n-1}) = (-1)^{|b|}\mu_{1;n}(b; a_1, \dots, a_n; \lambda_1, \dots, \lambda_{n-1})$ on $n-1$ open-sector spectral variables. □

Remark 8.2 (The Swiss-cheese pair is the explanatory mechanism; the bar is a computational model). The 2-real to 3-real dimensional uplift is the universal normal-collar direction of the codimension-one boundary condition forced by the chiral Deligne–Tamarkin terminal-object construction of Theorem 8.1. It is not produced by iterating the bar functor on $\mathcal{A}$ alone. The bar coalgebra $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is E_1-coassociative on the curve geometry: a single twisting coalgebra over the chiral associative operad $(\mathrm{ChirAss})^!$, with deconcatenation coproduct and bar differential constituting one graded coalgebra structure –- not two colours of a Swiss-cheese algebra. The bar serves as a chain-level computational model that resolves $\mathcal{A}$ for the purpose of computing the closed-colour Hochschild cochain complex via

$$Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) \;\simeq\; \mathrm{Hom}_{\mathrm{ch}}\big(\overline{B}^{\mathrm{ord}}(\mathcal{A}),\ \mathcal{A}\big).$$

The averaging map av_R of Section 4 acts within the open colour, projecting the ordered bar onto the symmetric bar; it does not construct or witness the closed colour. The open and closed colours are linked only through the Swiss-cheese mixed operations $\mu_{p;q}$.

In particular, the slogans *"$\overline{B}^{\mathrm{ord}}(\mathcal{A})$ is the bulk"*, *"the bar direction explains the 2d $\to$ 3d HT lift"*, and *"the bar is the open–closed transition"* are category-level errors: the universal closed sector is the closed colour $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$, the dimensional uplift is the universal collar of the Swiss-cheese pair, and the open–closed transition is the chiral Deligne–Tamarkin morphism, not the bar functor.

Remark 8.3 (Where the dimensional uplift comes from in $\mathsf{SC}^{\mathrm{ch,top}}$). The chiral Swiss-cheese chain-level construction adapts the topological Voronov pair [17] to the chiral setting: the open colour records E_1-chiral operations on $\mathcal{A}$ along the curve X (one complex direction, two real directions); the closed colour records E_2-topological brace operations on $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ in the transverse half-plane (two real directions). The chiral Deligne–Tamarkin terminal-object construction supplies a universal morphism between these two sectors that is itself a chiral object, with the inner conformal vector of the algebra controlling $\bar{Q}$-translations on the topological collar. The total $\mathsf{SC}^{\mathrm{ch,top}}$ structure is built from a holomorphic chiral colour on the boundary curve and a topological collar normal to it; the dimensional count is $2 + 1 = 3$. This mechanism is neither the Costello–Gwilliam factorisation construction, which is single-colour ambient, nor Voronov's topological Swiss-cheese alone, which lacks a holomorphic colour –- it is their fusion via chiral Deligne–Tamarkin. The bar of E_1 is again E_1 (cofree conilpotent over T^c); the Swiss-cheese pair (E_2, E_1) at the chiral-topological level is the source of the dimensional uplift.

9. Conclusion and open problems

The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, equipped with its deconcatenation coproduct, is the level-2 chain object before averaging in modular Koszul duality, and the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ is its Σ_n-coinvariant shadow under an admissible averaging comparison av. The five foundational theorems A, B, C, D, H of modular Koszul duality admit ordered E_1 formulations on the Koszul/descent locus. Theorem D is the degree-two calculation: the scalar degree-2 shadow is recovered from the spectral r-matrix $r(z)$ by averaging, giving κ for abelian families and κ_{dp} for non-abelian affine Kac–Moody before the Sugawara shift $\dim(\mathfrak{g})/2$ is added; the passage from $r(z)$ to the scalar shadow is a strict projection with nontrivial kernel.

Several problems remain open.

The full E_1 Koszul duality theorem. Theorems 5.1 and 5.2 prove the FAss-coalgebra structure, the inversion quasi-isomorphism on the Koszul locus, and compatibility with averaging. They do not prove an unconditional equivalence between all E_1-chiral algebras and cocomplete FAss-coalgebras. A full equivalence of categories requires tighter control of the coderived category of FAss-coalgebras and a characterisation of the Koszul locus that is independent of the commutative shadow.

The line-side modular R-matrix at higher genus. Theorem 5.4 packages the ordered degree-2 shadow into a formal series $R^{E_1,\mathrm{bin}}(z;\quad)$ whose g-th coefficient is the ordered degree-2 correction at genus g. Whether this formal series has a line-side interpretation as a genuine higher-genus modular R-matrix is a separate question. Ribbon-graph physics predicts such an object from the 't Hooft expansion, and the cyclic trace formalism gives the combinatorial language. Beyond genus one this remains an open construction.

GRT_1**-torsor structure of splittings.** Proposition 4.6 isolates the point at which aritywise linear splittings cease to be canonical: compatibility with the Maurer–Cartan equation depends on associator data. The Etingof–Kazhdan theorem provides this structure for Lie bialgebras. The chiral analogue, including a functorial lift $\kappa \mapsto r(z)$, remains open.

Swiss-cheese Koszul duality and the brane sector. Theorem 8.1 supplies the universal local Swiss-cheese pair $(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ as a terminal object, identifying the bulk and the open–closed mechanism that lifts 2d chiral data to a 3d holomorphic–topological theory. The averaging map of this paper is internal to the open colour and does not extend to a $\mathsf{SC}^{\mathrm{ch,top}}$-functor. A Koszul duality theorem for the chiral Swiss-cheese operad itself, that would package the two-coloured operad together with its bar / cobar adjunction and bind the brane / boundary-condition data of Volume II to the terminal pair, remains open. The mixed sector governing brane boundaries and the open–closed coupling belongs to Volume II.

In modular Koszul duality, the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is the primitive object and the symmetric bar is its R-twisted coinvariant shadow when descent exists. The five theorem surfaces A, B, C, D, H are the invariants visible after averaging. At degree 2, Heisenberg satisfies $\mathrm{av}_R(r(z)) = \kappa(\mathcal{A})$; non-abelian affine Kac–Moody satisfies $\mathrm{av}_R(r(z)) = \kappa_{\mathrm{dp}}(\mathcal{A})$, with the Sugawara shift $\dim(\mathfrak{g})/2$ supplying the full $\kappa(\mathcal{A})$.

Appendix A. Ordered-locus evidence

The ordered comparison becomes a theorem on the locus where the bar construction, the R-twisted descent, and the associator data are all present.

Definition A.1 (Ordered Koszul chiral algebra)**.** A chiral algebra $\mathcal{A}$ on a smooth curve X is *ordered Koszul* if:

(a) the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct carries a coalgebra deformation retract onto its cohomology over $(\mathrm{ChirAss})^!$;

(b) the counit $\Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism in the Francis–Gaitsgory factorization ambient;

(c) the R-twisted descent from ordered to symmetric bar is compatible with the chiral twisting morphism. This requires the Yang–Baxter equation and the unitarity relation $R(z)R^{\mathrm{op}}(-z) = \mathrm{id}$ after the scalar normalization appropriate to the family.

Proposition A.2 (Yangian ordered Koszul instance)**.** *The chiral Yangian $Y(\mathfrak{g})^{\mathrm{ch}}$, defined from the Drinfeld second presentation and Etingof–Kazhdan flatness, is ordered Koszul in the sense of Definition* A.1.

Proof. The Drinfeld second-presentation generators assemble an ordered bar complex controlled by pure-braid Orlik–Solomon cohomology. The Shelton–Yuzvinsky Koszulity theorem [14] gives the required quadratic Koszul calculation on the pure-braid algebra, while Etingof–Kazhdan flatness supplies the chiral twisting morphism and the normalized R-twisted descent. □

Theorem A.3 (Ordered quantum-group comparison on the Koszul locus)**.** *On the finite-type strongly admissible E_1 Koszul locus, after fixing a Drinfeld associator Φ and a strong-unitary R-twisted descent gauge, the following structures are equivalent descriptions of the same ordered bar coalgebra:*

(i) *the classical R-matrix $r_{\mathcal{A}}(z)$;*

(ii) *the A_∞-structure transferred to $\overline{B}^{\mathrm{ord}}(\mathcal{A})$;*

(iii) *the deconcatenation coproduct on $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ with its compatible chiral twisting morphism, coassociative up to Φ.*

The asserted H^0 associator-independence for the $\mathfrak{sl}_2$ Yangian remains a separate first-order computation until it is inscribed in the body theorem.

Proof. A choice of Drinfeld associator identifies the ordered tensor calculus with the quasi-triangular deformation supplied by Etingof–Kazhdan. The transferred A_∞ operations are then read from the same associator-controlled Maurer–Cartan element as the R-matrix. Conversely, the twisting morphism and deconcatenation coproduct recover the binary R-matrix and its higher associator corrections. □

Remark A.4 (Further ordered-locus extensions). Three extensions remain conditional. For $\mathfrak{gl}_N$, the chiral Drinfeld double can be constructed internally from Miura data, cross-arity Stasheff operations, and $U(\widehat{\mathfrak{gl}}_1)$ screenings. For $\mathcal{W}_\infty$, the ordered comparison belongs to the weight-completed class-M category. In genus one, Felder's dynamical $R(z,\tau)$ and the Fay trisecant identity [6, 7] replace the pentagon identity; higher genus requires theta-divisor input.

Remark A.5 (Toroidal formal-disk comparison). The toroidal formal-disk comparison uses Schiffmann–Vasserot CoHA [13] and the Ding–Iohara–Miki presentation [8] of $U_{q,t}(\ddot{\mathfrak{sl}}_N)$ with spectral parameters (u,v). It is a formal-disk statement. The global $\mathbb{P}^1 \times \mathbb{P}^1$ comparison requires the class-M chain-level topologization of the original bar complex.

Remark A.6 (Degree-two constants). The degree-two averaging constants are:

- Heisenberg $\mathcal{H}_k$: $\mathrm{av}_R(r(z)) = \kappa(\mathcal{H}_k) = k$.
- Non-abelian affine Kac–Moody $V_k(\mathfrak{g})$: $\mathrm{av}_R(r(z)) = \kappa_{\mathrm{dp}}(V_k(\mathfrak{g})) = k\dim(\mathfrak{g})/(2h^\vee)$, and $\kappa(V_k(\mathfrak{g})) = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$ after the Sugawara shift $\dim(\mathfrak{g})/2$.
- Virasoro Vir_c: $r(z) = (c/2)/z^3 + 2T/z$, and $\kappa(\mathrm{Vir}_c) = c/2$.
- Principal $\mathcal{W}_N$: $\kappa(\mathcal{W}_N) = c(H_N - 1)$ with $H_N = \sum_{j=1}^N 1/j$; for $N = 2$ this recovers $\kappa(\mathrm{Vir}_c) = c/2$.

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